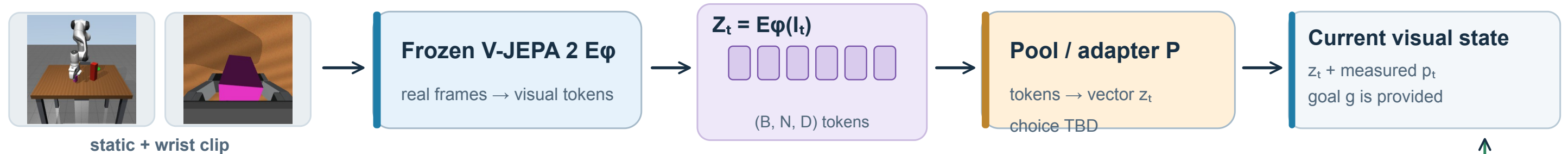


Inference: observe → imagine → act → replan

PROPOSED SYSTEM

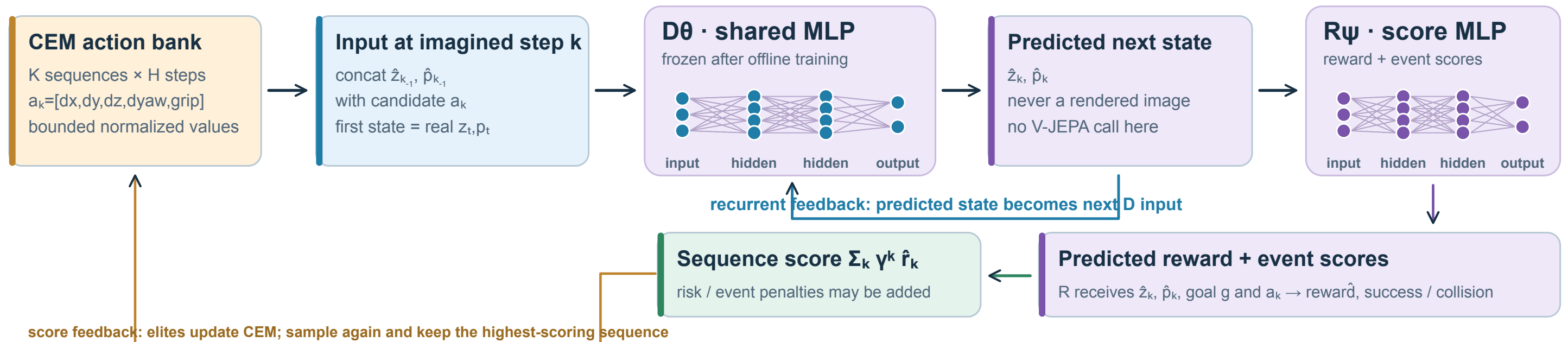
A frozen visual encoder grounds the current state; learned D and R evaluate imagined action sequences

01 OBSERVE REALITY · V-JEPA RUNS ON NEW CAMERA CLIPS



p_t comes from the robot/controller sensors. Object coordinates are not read from privileged MuJoCo state by this proposed vision pipeline.

02 PLAN IN LATENT SPACE · SAMPLE SEQUENCES, RECURRENTLY APPLY D, SCORE WITH R



Example candidate actions: $[0, 0, -0.5, 0, +1]$ asks for 2 cm downward with gripper open; $[0.6, 0, 0, 0, -1]$ asks to move sideways and close. Reward examples are predictions, not ground-truth.

03 EXECUTE ONE ACTION IN THE REAL SIMULATOR · GET NEW EVIDENCE



Closed loop: real frames → $E\phi$ → z_t → CEM with D/R → one real action → new frames. D and R are proposed; current repo still runs the scripted policy.